Strong CP without axions from the center of $\mathrm{SU}(2)_R$

Dino Škrgić⋆ and Dejan Stojkovic†

Department of Physics, University at Buffalo, State University of New York, Buffalo, NY 14260, USA

⋆ dinoskrg@buffalo.edu , † ds77@buffalo.edu

Abstract

Nelson–Barr solutions of the strong CP problem hinge on a single missing entry in the quark mass matrix, and on a symmetry good enough to keep it missing in the presence of gravity. We show that the Standard-Model hypercharges already point to such a symmetry. They arrange the right-handed fermions into doublets of an $\mathrm{SU}(2)_R$, and if the vector-like quark of Nelson–Barr is an $\mathrm{SU}(2)_R$ singlet, the dangerous coupling to the Higgs bidoublet has no invariant at all. The selection rule is the center of $\mathrm{SU}(2)_R$; a triplet breaking $\mathrm{SU}(2)_R$ leaves it intact as a discrete gauge symmetry, which is matter parity in disguise. With CP broken spontaneously by two $\mathrm{SU}(2)_R$ doublets, $\arg\det\mathcal{M}$ vanishes for any Yukawa texture while the CKM phase is generically of order one, and Planck-suppressed corrections first appear at dimension five, placing the CP-breaking scale below $\sim 10^9$ GeV, a bound the bidoublet vacuum reproduces once the potential is minimized. In the context of the recently proposed model by Arkani-Hamed et al. [1], drawn as a link diagram, the forbidden coupling isolates one node, so no perfect matching of the graph can pass through a complex link, and the one physical phase becomes the relative phase of two vacuum expectation values. An exhaustive scan of the 6696 eleven-link graphs shows that every texture compatible with the measured unitarity angles locks α to this phase; at $\pi/2$, the value a $\mathbb{Z}_4$ exchange symmetry of the potential selects, they give $(\alpha, \beta, \gamma) \simeq (90°, 23.9°, 66°)$, with β fixed by $|V_{ub}/V_{cb}|$ through $\sin\beta = R_b \sin\alpha$.

Copyright attribution to authors.
This work is a submission to SciPost Physics.
License information to appear upon publication.
Publication information to appear upon publication.

Received Date
Accepted Date
Published Date

Contents

1 Introduction 2

2 Centers as selection rules 3

3 The right-handed doublets and their center 4

4 A gauge origin for the Nelson–Barr zero 5
4.1 The CP-breaking sector . . . 5
4.2 The texture and the determinant . . . 5
4.3 The bidoublet vacuum . . . 6
4.4 Below the SU(2)$_R$ scale: matter parity and its quality . . . 8

5 The unitarity triangle 9
5.1 Link diagrams and perfect matchings . . . 9
5.2 The minimal graph . . . 10
5.3 All eleven-link graphs . . . 11
5.4 Without the vector-like quark . . . 12
5.5 Why $\pi/2$, and why not $\pi/8$ . . . 12
5.6 An axion instead? . . . 14

6 Conclusions 14

A Registers in components 15

B Numerical procedure 15

1 Introduction

The QCD vacuum angle enters physics only through the combination

$$\bar{\theta} = \theta_{\text{bare}} + \arg\det(M_u M_d), \tag{1}$$

which ties a property of the gauge sector to the phases of the quark mass matrices. The neutron electric dipole moment bounds it by $|\bar{\theta}| \lesssim 10^{-10}$ [2], and nothing in the Standard Model (SM) explains why. An axion [3,4] relaxes $\bar{\theta}$ dynamically, but needs a global U(1)$_{\text{PQ}}$ that gravity must respect to extraordinary accuracy [5–7]. Nelson and Barr [8,9] took the other road: impose CP in the ultraviolet, so that $\theta_{\text{bare}} = 0$, and break it spontaneously in a way that leaves the determinant real while the CKM matrix is not. The minimal realization, due to Bento, Branco and Parada [10], adds one vector-like quark and needs one block of the mass matrix to vanish. That zero is the whole mechanism, and it is usually enforced by a $\mathbb{Z}_2$ introduced for the purpose. A global $\mathbb{Z}_2$ is exposed to the same Planck-suppressed operators that trouble the axion [11], which has motivated constructions in which the zero follows from gauge symmetry [12–15], alongside parity solutions in left-right models [16,17] and gauged discrete P or CP [18].

The point of this paper is that the SM spectrum already contains a gauge origin for the zero. The hypercharges of the right-handed fermions are exactly those of doublets of an SU(2)$_R$ on a $B-L$ background [19–21]. A vector-like quark D with the SM charges of d_R but no SU(2)$_R$ index cannot couple to the quark doublets through the Higgs bidoublet, because $\bar{q}_L \Phi D_R$ is an SU(2)$_R$ doublet. What forbids it is the center of SU(2)$_R$, and a triplet that breaks SU(2)$_R$ leaves the center unbroken. Below the breaking scale the zero is therefore protected by a discrete gauge symmetry in the sense of Krauss and Wilczek [22,23]; up to unbroken gauge transformations it is the matter parity $(-1)^{3(B-L)}$ familiar from left-right and $B-L$ models [24,25]. None of these ingredients is exotic. Put together, they give a Nelson–Barr model whose texture zero, CP-breaking scalars

and seesaw scale all come from the left-right gauge structure, and whose quality can be estimated operator by operator.

The same zero has a striking consequence for flavor. Arkani-Hamed, Figueiredo, Hall and Manzari [1] observed that the unitarity triangle sits close to $\alpha = \pi/2$, $\beta = \pi/8$, $\gamma = 3\pi/8$, and organized the Yukawa sector as link diagrams, bipartite graphs whose nodes are chiral fields and whose links are nonzero Yukawa entries. In that language the determinant is a sum over perfect matchings of the graph. The $\mathrm{SU}(2)_R$-forbidden links leave the vector-like node with a single neighbor, and as a result no matching can contain a complex link. The graph also locates the one physical phase: it is the relative phase of two vacuum expectation values (VEVs). We enumerate every graph with this field content and a single phase, fit each to the quark masses and mixing magnitudes, and find that the textures surviving the measured angles all tie α to that phase.

This work began in a program that rebuilds quantum kinematics and relativity from the counting statistics of discrete sequences [26–32], where internal symmetries appear as relabelings of finite alphabets and their centers are the part of the symmetry that survives coarse graining. That viewpoint is what drew our attention to the center of $\mathrm{SU}(2)_R$, and Sec. 2 uses its vocabulary sparingly. Every result below is stated and derived in ordinary field theory.

2 Centers as selection rules

Label the internal state of a field by its indices. An N-valued index lives in a space $\mathcal{H}_N$ spanned by $|1\rangle, \ldots, |N\rangle$, which we call a register, and its norm-preserving relabelings form $\mathrm{U}(N)$, the product of $\mathrm{SU}(N)$ and a phase up to a common $\mathbb{Z}_N$. A one-valued register carries only a phase, $e^{in_1\alpha}$ for n_1 indices; a two-valued one carries weak isospin, generated by $\sigma_1 = e_{10} + e_{01}$, $\sigma_2 = i(e_{01} - e_{10})$ and $\sigma_3 = e_{11} - e_{00}$ in the matrix units $e_{ij}|k\rangle = \delta_{jk}|i\rangle$; a three-valued one carries color, with the six off-diagonal e_{ij} and the two diagonal combinations $\lambda_{3,8}$ as the eight gluons (Appendix A). Keeping the phase of the first and the $\mathrm{SU}(N)$ parts of the other two gives $\mathrm{U}(1)_Y \times \mathrm{SU}(2)_L \times \mathrm{SU}(3)_C$, with $1+3+8$ adjoint gauge bosons that carry no net index number in any register. The register sizes are of course read off from nature, and so is the choice to drop two of the three phases; what the language buys is a clean view of the centers.

The center $\mathbb{Z}_N \subset \mathrm{SU}(N)$ [33] multiplies every fundamental index by $\omega = e^{2\pi i/N}$ and every antifundamental index by ω^{-1}, whatever their values. For color, a state with a quarks and $\bar{a}$ antiquarks picks up ω^t with triality $t = (a - \bar{a}) \bmod 3$, the label of a character of $\mathbb{Z}_3$ in the same way that q labels the character $e^{iq\alpha}$ of $\mathrm{U}(1)$. Triality adds under tensor products, is constant on an irreducible representation, $t = (p - q) \bmod 3$ for Dynkin labels (p, q) (Table 1), and is conserved because gauge bosons have $t = 0$. Singlets have $t = 0$, but not conversely: the **8**, **10** and **27** are triality-neutral too. The baryon $\varepsilon_{ijk} q^i q^j q^k$ is invariant because $\det U = 1$, which for $U = \omega \mathbf{1}$ reads $\omega^3 = 1$. The role of $\mathbb{Z}_N$ in confinement, through Polyakov loops, center vortices and N-ality dependent string tensions, is well established [34–36]; here color serves only as the familiar template for what follows.

Triality also fixes the unit of hypercharge. If every color singlet is to carry integer charge, one color index must carry charge in units of $1/N_c$, so with $Q = T_{3L} + Y/2$

$$Y = \frac{n_1}{3}, \qquad n_1 \in \mathbb{Z}, \tag{2}$$

an input equivalent to the $\mathbb{Z}_6$ quotient of the global SM gauge group [37]. A finite register is a finite-dimensional system, but $\mathrm{SU}(N)$ and its center act on $\mathcal{H}_N^{\otimes n}$ exactly for every n; finiteness affects spectra, not selection rules [38].

Table 1: Triality of low-lying SU(3) representations.

irrep	**1**	**3**	$\bar{\mathbf{3}}$	**6**	**8**	**10**
(p,q)	$(0,0)$	$(1,0)$	$(0,1)$	$(2,0)$	$(1,1)$	$(3,0)$
t	0	1	2	2	0	0

Table 2: One SM generation, with $n_1 \equiv 3Y$, $T_{3L} = b - \frac{1}{2}$ ($b \in \{0,1\}$) and T_{3R} from Eq. (3). Every row satisfies Eq. (4). The bidoublet $\Phi \sim (\mathbf{2},\mathbf{2})$ of $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ has $B-L=0$; the half listed has $T_{3R} = +\frac{1}{2}$ and $Y = +1$, the other half $T_{3R} = -\frac{1}{2}$ and $Y = -1$.

field	n_1	Y	T_{3L}	T_{3R}	$B-L$	Q
$u_L,\ d_L$	$+1$	$+\frac{1}{3}$	$\pm\frac{1}{2}$	0	$+\frac{1}{3}$	$+\frac{2}{3},\ -\frac{1}{3}$
u_R	$+4$	$+\frac{4}{3}$	0	$+\frac{1}{2}$	$+\frac{1}{3}$	$+\frac{2}{3}$
d_R	-2	$-\frac{2}{3}$	0	$-\frac{1}{2}$	$+\frac{1}{3}$	$-\frac{1}{3}$
$\nu_L,\ e_L$	-3	-1	$\pm\frac{1}{2}$	0	-1	$0,\ -1$
ν_R	0	0	0	$+\frac{1}{2}$	-1	0
e_R	-6	-2	0	$-\frac{1}{2}$	-1	-1
Φ^+, Φ^0	$+3$	$+1$	$\pm\frac{1}{2}$	$+\frac{1}{2}$	0	$+1,\ 0$

3 The right-handed doublets and their center

Table 2 lists one generation with the integer $n_1 \equiv 3Y$ and $B-L = +\frac{1}{3}$ for quarks, -1 for leptons. The right-handed neutrino carries nothing at all, yet it is required as soon as $B-L$ is gauged: $(B-L)^3$ and $B-L$ summed over the left-handed Weyl fermions of a generation give -1 without it and zero with it. Define

$$T_{3R} \equiv \tfrac{1}{2}\big[Y - (B-L)\big]. \tag{3}$$

On the observed spectrum it takes only the values $0, \pm\frac{1}{2}$; the states with $|T_{3R}| = \frac{1}{2}$ pair into (u_R, d_R) and (ν_R, e_R) with equal $B-L$ and color; and every field satisfies

$$n_1 = 6\,T_{3R} + 3\,(B-L). \tag{4}$$

This is the observation on which left-right symmetric models are built [19–21]: the hypercharges are those of

$$G_{\mathrm{LR}} = \mathrm{U}(1)_{B-L} \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R \times \mathrm{SU}(3)_C, \tag{5}$$

with $Y = 2T_{3R} + (B-L)$. The bookkeeping closes on the bosons as well (Table 3): all unbroken gauge bosons have $n_1 = 0$, and W_R^+, which must turn d_R into u_R and e_R into ν_R, carries $n_1 = 6$, which is what both transitions require, $4-(-2) = 0-(-6)$. We take G_{LR} with its bidoublet Φ, its triplet Δ_R and the seesaw as given, and ask about the one object the left-right literature rarely looks at: the center of $\mathrm{SU}(2)_R$.

The center of $\mathrm{SU}(2)_R$ is $\{\mathbf{1}, -\mathbf{1}\}$, and its character is

$$P_R = (-1)^{2T_{3R}}, \tag{6}$$

Table 3: Bosons of the model. χ_R are the CP-breaking doublets of Sec. 4.1, and P_R is the center parity of Eq. (6).

field	T_{3L}	T_{3R}	$B-L$	n_1	Y	P_R
$B_\mu,\ W_L,\ G_\mu$	$0, \pm 1$	0	0	0	0	$+$
$W_R^\pm,\ W_R^3$	0	$\pm 1,\ 0$	0	$\pm 6,\ 0$	$\pm 2,\ 0$	$+$
Φ^0	$-\frac{1}{2}$	$+\frac{1}{2}$	0	3	$+1$	$-$
χ_R^0	0	$+\frac{1}{2}$	-1	0	0	$-$
Δ_R^0	0	-1	$+2$	0	0	$+$

the number of $\mathrm{SU}(2)_R$ indices modulo two. It is even on q_L, L_L, Δ_R and all gauge bosons, and odd on u_R, d_R, ν_R, e_R, Φ and χ_R. Because the center sits inside the gauge group, every $\mathrm{SU}(2)_R$ invariant is center-even: a product of isospins $j_1, \ldots, j_k$ contains a singlet only if $\sum_i j_i$ is an integer. An operator with an odd number of $\mathrm{SU}(2)_R$ doublets therefore never appears, at any order, and this can be read off by counting.

The parity tells apart fields that the SM cannot. An $\mathrm{SU}(2)_R$-singlet quark D with the hypercharge of d_R must have $B-L=-\frac{2}{3}$:

$$d_R:\ (T_{3R}, B-L, P_R) = (-\tfrac{1}{2}, +\tfrac{1}{3}, -), \qquad D_R:\ (0, -\tfrac{2}{3}, +), \tag{7}$$

with identical color, T_{3L} and Y. The SM Yukawa $\bar{q}_L \Phi Q_R$ contracts one index of each SU(2) through the bidoublet, so a chirality flip moves a quark from an $\mathrm{SU}(2)_R$ doublet into an $\mathrm{SU}(2)_L$ one. Masses, mixings and $\bar{\theta}$ are all governed by which such moves gauge invariance allows.

4 A gauge origin for the Nelson–Barr zero

4.1 The CP-breaking sector

We take the ultraviolet theory to be CP invariant, so that all couplings are real and $\theta_{\mathrm{bare}} = 0$, and let CP be broken spontaneously by scalars $\chi_R^{(a)}$ that are $\mathrm{SU}(2)_R$ doublets with $B-L=-1$. Their upper components are neutral ($Y = 1-1 = 0$) and P_R-odd. One such doublet is not enough, since its phase can be rotated away and the CKM matrix comes out real; with two, the relative phase of $\langle \chi_R^{(1)} \rangle$ and $\langle \chi_R^{(2)} \rangle$ is physical [39]. The vector-like quark $D_{L,R}$ of Sec. 3 completes the field content; its Dirac mass $\mu \bar{D}_L D_R$ is P_R-even and real.

4.2 The texture and the determinant

Table 4 lists what the parity allows. In the basis (d_R, D_R) against $(\bar{d}_L, \bar{D}_L)$ the down-type mass matrix takes the Bento–Branco–Parada form [10],

$$\mathcal{M} = \begin{pmatrix} m & 0 \\ B & \mu \end{pmatrix}, \qquad B = \sum_a F^{(a)} \langle \chi_R^{(a)} \rangle \in \mathbb{C}^{1\times 3}, \tag{8}$$

with $m = Y_d v/\sqrt{2}$ real (the bidoublet vacuum is relatively real to the required accuracy, Sec. 4.3) and the zero standing for the absent $\bar{q}_L \Phi D_R$. Expanding along the last column,

$$\det \mathcal{M} = \mu \, \det m, \tag{9}$$

Table 4: Operators of the down sector and their center parity. A P_R-odd operator contains an odd number of $\mathrm{SU}(2)_R$ doublets and has no invariant. Here d_R stands for the Q_R doublet, and $\mathrm{SU}(2)_R$ indices are contracted in the unique invariant way. The last three rows are the leading Planck-suppressed corrections of Sec. 4.4.

operator	factor parities	P_R	role
$\bar{q}_L \Phi d_R$, $\bar{L}_L \Phi e_R$	$(+)(-)(-)$	$+$	SM Yukawas
$\bar{\nu}_R^c \nu_R \Delta_R$	$(-)(-)(+)$	$+$	seesaw
$\mu\, \bar{D}_L D_R$	$(+)(+)$	$+$	Dirac mass, real
$\bar{D}_L\, \chi_R^{(a)}\, d_R$	$(+)(-)(-)$	$+$	bridge B, complex
$\bar{q}_L \Phi D_R$	$(+)(-)(+)$	$-$	**absent**
$\bar{q}_L \Phi \chi_R^\dagger D_R / M_{\mathrm{Pl}}$	$(+)(-)(-)(+)$	$+$	dimension 5
$\bar{q}_L \Phi\, \chi_R\, \Delta_R\, D_R / M_{\mathrm{Pl}}^2$	$(+)(-)(-)(+)(+)$	$+$	dimension 6
$\bar{q}_L \Phi \chi_R^\dagger \lvert\chi_R\rvert^2 D_R / M_{\mathrm{Pl}}^3$	$(+)(-)(-)(-)(-)(+)$	$+$	dimension 7

so the complex bridge drops out and $\bar{\theta}_{\mathrm{tree}} = 0$, as an identity rather than a tuning. (Reality fixes the determinant's phase modulo π; we work in the convention with positive quark masses, so the CP-symmetric vacuum sits at $\bar{\theta} = 0$.) The phases survive in mixing, which is controlled by the Hermitian square. Integrating out D gives, to leading order in $mB^\dagger/(|B|^2 + \mu^2)$,

$$H_{\mathrm{eff}} = mm^\dagger - \frac{m\, B^\dagger B\, m^\dagger}{|B|^2 + \mu^2}, \tag{10}$$

whose second term is complex through the relative phases of the B_i and produces an order-one CKM phase when $|B| \sim \mu$.

To see both effects at once we drew 400 real textures $m_{ij} = c_{ij}\lambda^{a_i + a_j}$, with $\lambda = 0.225$, charges $(2, 1, 0)$ for the down sector and $(4, 2, 0)$ for the up sector, random signs and $|c_{ij}|$ log-uniform in $[\frac{1}{2}, 2]$, normalized to $m_b = 2.9$ GeV and $m_t = 173$ GeV, together with a random complex bridge of size $|B| = \mu$ (Appendix B). The textures reproduce the quark spectrum only to within factors of a few, by construction; the CP violation is not put in. The determinant is real to $|\operatorname{Im}\det\mathcal{M}|/|\det\mathcal{M}| < 2 \times 10^{-13}$ in every draw, while $|J|$ has median 7.4×10^{-5} and 10–90% range 5×10^{-6} to 7×10^{-4}, with the measured $J = 3.08 \times 10^{-5}$ [40] at the 30th percentile (Fig. 1). With a single doublet, $|J|$ is zero to numerical precision. Restoring the forbidden links at $\epsilon = 10^{-6}$ moves $\bar{\theta}_{\mathrm{tree}}$ to $\sim \epsilon$, which is the size of the protection the zero provides.

4.3 The bidoublet vacuum

The bidoublet has two neutral VEVs, κ_1 and $\kappa_2 e^{i\alpha}$, and gives $M_u = \kappa_1(h + \tilde{h}\,\bar{z})$ and $M_d = \kappa_1(hz + \tilde{h})$ with $z \equiv (\kappa_2/\kappa_1)e^{i\alpha}$, real $h, \tilde{h}$ multiplying Φ and $\tilde{\Phi} = \sigma_2 \Phi^* \sigma_2$. A relative phase feeds Eq. (1) directly: to first order in $y \equiv \operatorname{Im} z$, and writing $x \equiv \Re z$,

$$\bar{\theta}_{\mathrm{tree}} = c\, \operatorname{Im} z, \qquad c = \frac{\mathrm{Tr}(M_d^{-1} M_u) - \mathrm{Tr}(M_u^{-1} M_d)}{1 - x^2}, \tag{11}$$

which for the random textures of Sec. 4.2 has median $|c| \simeq 45$ at $x = 0$ and a 10–90% range of 9 to 300, dominated by m_t/m_b. So the question is what fixes $\operatorname{Im} z$, and the answer comes from minimizing the potential.

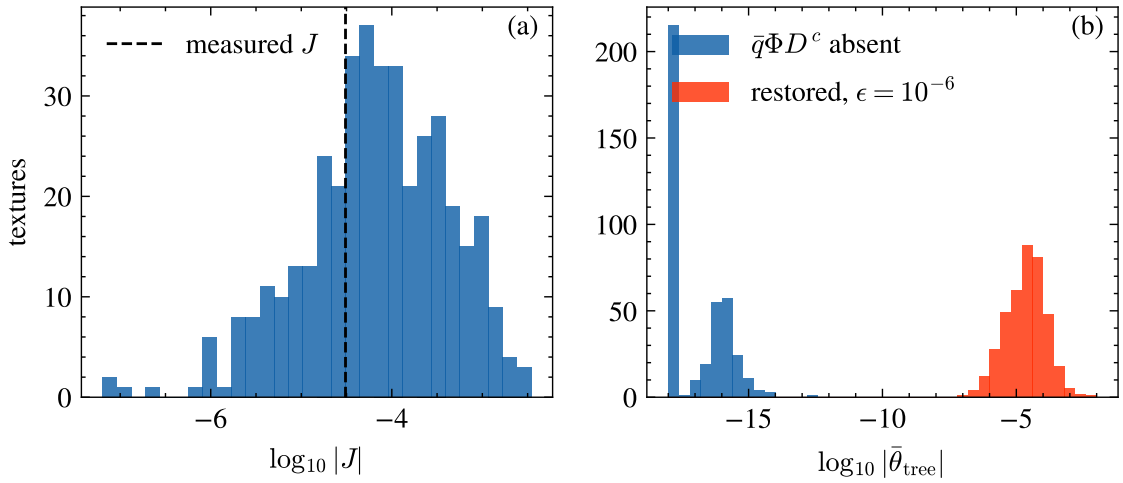


Figure 1: (a) $|J|$ for 400 random Froggatt–Nielsen textures with the texture of Eq. (8) and two CP-breaking doublets; the measured value is dashed. (b) The tree-level $\bar{\theta}$ of the same textures, at machine precision, and with the forbidden links q_i–D^c restored at $\epsilon = 10^{-6}$.

With all couplings real, every term linear in $\mathrm{Tr}(\Phi^\dagger\tilde{\Phi})$ has a real coefficient—μ_2^2, $\lambda_4\,\mathrm{Tr}(\Phi^\dagger\Phi)$, $\alpha_2\,\mathrm{Tr}(\Delta_R^\dagger\Delta_R)$ and the diagonal portals $g_{aa}\,\chi_a^\dagger\chi_a$—except the off-diagonal portal to the two CP-breaking doublets,

$$g_{ab}\,\chi_R^{(a)\dagger}\big(\Phi^\dagger\tilde{\Phi}\big)\chi_R^{(b)} + \text{h.c.}, \qquad g_{ab}\in\mathbb{R}, \tag{12}$$

whose vacuum value is $2\kappa_1\kappa_2\,g_{ab}w_aw_b\cos(\alpha+\theta_a-\theta_b)$. Summed over a,b the symmetric part of g gives $4g^S\kappa_1\kappa_2w_1w_2\cos\alpha\cos\varphi$ and the antisymmetric part $g^A=\frac{1}{2}(g_{12}-g_{21})$ gives $4g^A\kappa_1\kappa_2w_1w_2\sin\alpha\sin\varphi$; the latter is the only term in the renormalizable potential that is odd in $\sin\alpha$. Writing $\kappa_2e^{i\alpha}=X+iY$, the potential is a quadratic form in (X,Y) plus quartics suppressed by κ^2/v_R^2, with Y entering only through $D\,Y^2+4g^Aw_1w_2\sin\varphi\,\kappa_1Y$, where $D\simeq\alpha_3v_R^2+\dots$ is the mass squared of the heavy doublet direction of the bidoublet. The minimum is therefore

$$\mathrm{Im}\,z=\frac{\kappa_2}{\kappa_1}\sin\alpha=-\frac{2g^A\,w_1w_2\sin\varphi}{D}, \qquad D=\mathcal{O}(\alpha_3v_R^2), \tag{13}$$

independently of μ_2^2, α_2 and λ_4, which only set $\Re z=\mathcal{O}(1)$. We verified Eq. (13) by minimizing the full potential at 60-digit precision, with the light doublet tuned to the electroweak scale and random order-one couplings: the numerical $\mathrm{Im}\,z$ agrees with Eq. (13) to twelve digits, and the bidoublet sector has no CP-violating minimum of its own unless D is tuned below $8|\lambda_2|\kappa_1\kappa_2$. Combining with Eq. (11),

$$\bar{\theta}_{\rm tree}\simeq\frac{2c\,g^A}{\alpha_3}\,\frac{\langle\chi_R\rangle^2}{v_R^2}\,\sin\varphi\;\approx\;10^2\,\frac{g^A}{\alpha_3}\Big(\frac{\langle\chi_R\rangle}{v_R}\Big)^2 \tag{14}$$

for $w_1=w_2=\langle\chi_R\rangle$ and $\varphi=\pi/2$. With $v_R\sim10^{15}$ GeV from the seesaw, $\bar{\theta}<10^{-10}$ requires $\langle\chi_R\rangle\lesssim10^9\,(\alpha_3/g^A)^{1/2}$ GeV—the same window as Eq. (18). The bidoublet is thus relatively real to the required accuracy because the CP-breaking scale lies six orders of magnitude below the $\mathrm{SU}(2)_R$ scale; no additional assumption is needed, and the hierarchy between the two scales, which the seesaw and the quality bound each impose separately, is now also required by the bidoublet. If the scalar potential is symmetric under the exchange of the two doublets (Sec. 5.5), g^A vanishes at tree level and the residual phase is smaller still.

4.4 Below the $\mathrm{SU}(2)_R$ scale: matter parity and its quality

We break $\mathrm{SU}(2)_R \times \mathrm{U}(1)_{B-L}$ with a triplet Δ_R of $B-L=+2$, whose neutral component has $T_{3R} = -1$ and $Y = 0$. Integer isospin makes it center-even, so $\langle \Delta_R \rangle$ leaves the center unbroken and the zero is protected below the breaking scale by a discrete gauge symmetry [22]. Writing the center element as $e^{2\pi i T_{3R}} = e^{i\pi Y} e^{-i\pi(B-L)}$ and composing with elements of the unbroken $\mathrm{U}(1)_Y$ and of the color center, one finds that on every field of the model

$$P_R \;\simeq\; (-1)^{3(B-L)}, \tag{15}$$

odd on quarks, leptons and χ_R, even on Φ, Δ_R and D. The residual symmetry is matter parity [24,25], and it is genuinely disconnected from the SM gauge group, since it is -1 on ν_R where $\mathrm{U}(1)_Y \times \mathrm{SU}(3)_C$ can only act trivially. In this form it is clear that the electroweak VEV ($B-L=0$) leaves it alone and that $\langle \chi_R \rangle$ ($B-L=-1$) breaks it, at the scale where CP is broken. The triplet doubles as the source of right-handed neutrino masses through $\bar{\nu}_R^c \nu_R \Delta_R$; with $m_\nu \simeq y_\nu^2 v^2/(2M_R)$ and $m_\nu \simeq 0.05$ eV, $M_R \sim 10^{15}$ GeV for $y_\nu = \mathcal{O}(1)$. No other $\mathbb{Z}_2$ built from the available charges, $(-1)^{2T_{3R}} f(B-L)$ with $f(0)=1$, can make the bidoublet even.

Gravity can still generate gauge-invariant operators that involve the condensates. We normalize them to $M_{\rm Pl} = 1.22\times10^{19}$ GeV; a mediator of mass $\Lambda < M_{\rm Pl}$ tightens everything below by $M_{\rm Pl}/\Lambda$. The leading one is

$$\mathcal{O}_5 = \frac{\bar{q}_L \, \Phi \, \chi_R^{(a)\dagger} \, D_R}{M_{\rm Pl}}, \tag{16}$$

$B-L$ neutral and with two $\mathrm{SU}(2)_R$ doublets; its contribution to $\bar{\theta}$ needs two doublets with different phases and vanishes for $n_\chi = 1$. Replacing $\chi_R^\dagger$ by $\chi_R^\dagger \chi_R$ violates $B-L$, but the triplet makes room for $\bar{q}_L \Phi \chi_R \Delta_R D_R / M_{\rm Pl}^2$ at dimension six, which feeds the same entry with an extra $\langle \Delta_R \rangle / M_{\rm Pl} \sim 10^{-4}$ and does not change the estimate. The dimension-five operator fills the empty block with $\Delta_{q_L D_R} \sim \langle \chi^{(a)} \rangle^* / M_{\rm Pl}$ times a flavor matrix, and through $\det \mathcal{M} = \det m \, (\mu - B m^{-1} \Delta)$ shifts

$$\delta\bar{\theta} \simeq \frac{\mathrm{Im}\left(B\, m^{-1}\Delta\right)}{\mu} \sim F \, \frac{\langle \chi_R \rangle^2}{M_{\rm Pl}\, \mu} \tag{17}$$

when the flavor matrix of $\mathcal{O}_5$ is aligned with the down Yukawas, so that $m^{-1}\Delta$ carries no hierarchy. Requiring $\delta\bar{\theta} < 10^{-10}$ with $|B| \sim \mu$, i.e. $\langle \chi_R \rangle \sim \mu/F$, gives

$$\langle \chi_R \rangle \sim \mu \lesssim 10^9 \text{ GeV}. \tag{18}$$

Figure 2 checks the scaling with the exact determinant. If the coefficients of $\mathcal{O}_5$ are flavor anarchic, the light-quark entries of m^{-1} enhance the leak by about two orders of magnitude, and if they are of order one rather than of the size of y_b, by a further $v/m_b \sim 60$, bringing the bound down to $10^{5\text{–}7}$ GeV. Equation (18) is thus the optimistic end of a range set by how much flavor structure the ultraviolet respects; it is still weaker than the $\sim 10^2$ TeV often quoted for Nelson–Barr models with $\Lambda \sim M_{\rm Pl}$.

With the exact texture the one-loop contribution to $\bar{\theta}$ vanishes [10]. The first loop effect is at two loops, governed by $\mathrm{Im}(B_i^* B_j)\,(mm^\dagger)_{ij}$ divided by μ^2, and is of order $(16\pi^2)^{-2} y_b^2 y_s^2$, or about 10^{-15} [11]; a complete calculation for this texture is left for future work. A vector-like quark at $10^{8\text{–}9}$ GeV has no observable flavor-changing effects: its right-handed mixing with light quarks is of order one, but the left-handed mixing is $\sim m_b |B|/(|B|^2+\mu^2) \lesssim 10^{-8}$. When $\langle \chi_R \rangle$ breaks the residual $\mathbb{Z}_2$ it makes domain walls, but since π_0 of the group left by $\langle \Delta_R \rangle$ is $\mathbb{Z}_2$, strings form first and the walls end on them [41]. Such networks collapse, and at $\langle \chi_R \rangle \sim 10^{8\text{–}9}$ GeV the walls never dominate the energy density; primordial black holes offer a further sink [42]. A quantitative treatment of the network is beyond our scope.

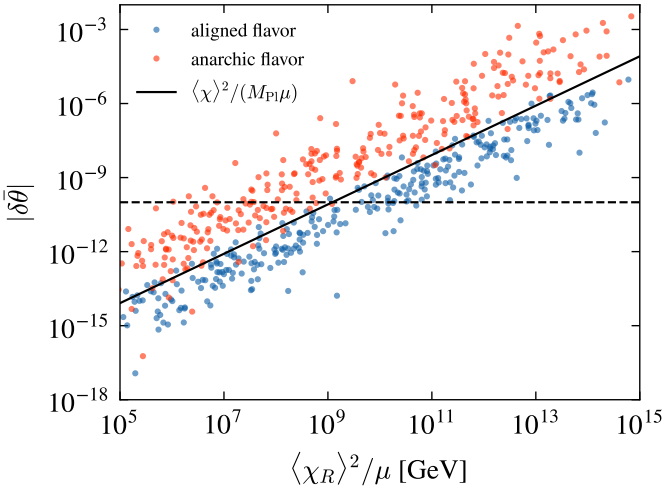


Figure 2: Tree-level $|\delta\bar\theta|$ from the dimension-five operator, computed from the exact determinant for random textures, bridges and scales, with flavor-aligned and anarchic coefficients. The solid line is $\langle\chi_R\rangle^2/(M_{\rm Pl}\mu)$ and the dashed line the bound 10^{-10}.

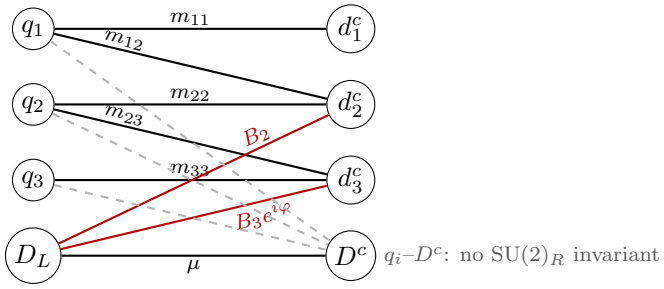


Figure 3: The down sector of the minimal graph of Sec. 5.2. Black links are real (Yukawas and the Dirac mass μ), red links are the complex bridges $B_j = \sum_a F^{(a)}\langle\chi_R^{(a)}\rangle$, and dashed links are forbidden by SU(2)$_R$. With D^c left a single neighbor, every perfect matching contains μ and none contains a bridge. The unique cycle $D_L \to d_2^c \to q_2 \to d_3^c \to D_L$ carries the phase $\varphi = \arg(B_2^* B_3)$.

5 The unitarity triangle

5.1 Link diagrams and perfect matchings

The measured angles [40, 43],

$$\alpha = (91.0 \pm 2.8)^\circ, \qquad \beta = (22.6 \pm 0.5)^\circ, \qquad \gamma = (66.4^{+2.7}_{-2.8})^\circ, \tag{19}$$

with α from unitarity, sit remarkably close to $\pi/2$, $\pi/8$ and $3\pi/8$ [1]. Following Ref. [1] we draw the Yukawa sector as a bipartite graph with left-handed nodes q_i, D_L and right-handed nodes u_j^c, d_j^c, D^c; links are nonzero entries and missing links are texture zeros (Fig. 3). One class of zeros, the links q_i–D^c, is now dictated by SU(2)$_R$.

For an $n \times n$ matrix, $\det Y = \sum_\pi \mathrm{sgn}(\pi) \prod_i Y_{i\pi(i)}$, and a permutation contributes only if each factor is a link, that is, if it is a perfect matching of the graph:

$$\det Y = \sum_{\text{perfect matchings } M} \pm \prod_{\ell \in M} Y_\ell . \tag{20}$$

Ref. [1] uses this through sparsity: a graph with a single perfect matching has a monomial determinant whose phase can be moved onto an unused link, which works for 94 of its 99 textures. Here the mechanism is different. A matching must cover D^c, whose only link is μ, so it uses D_L; every bridge ends on D_L, so no matching contains a bridge. Each term of Eq. (20) is then μ times a matching of the real m, which is Eq. (9) again, and it holds for any m, sparse or dense. Restoring one forbidden link at strength ϵ adds six phase-carrying matchings and an imaginary part of order ϵ (Fig. 1b).

The gauge zero protects $\bar{\theta}$; further zeros make the triangle predictive. A connected bipartite graph with V nodes and L links has $L - V + 1$ independent cycles, and since every node can absorb one phase, a single physical phase needs $L = V$. The nine-node graphs of Ref. [1] have nine links; ours have $V = 3+1+3+3+1 = 11$ and so eleven links, carrying eleven magnitudes and one phase. Only $|B|/\mu$ is visible at low energy, and if the two doublets, which carry identical charges, are exchanged by a symmetry of the scalar potential, then

$$|B_2| = |B_3|, \tag{21}$$

leaving ten magnitudes for ten observables and the phase as a prediction. On the cycle of Fig. 3 the rephasing invariant is

$$\varphi = \arg\big(m_{22} B_2^* B_3 m_{23}^*\big) = \arg\big(B_2^* B_3\big), \tag{22}$$

the relative phase of the two doublet VEVs. The "Yukawa triangle" of Ref. [1] is drawn by two VEVs: isosceles by Eq. (21), with apex angle φ, and its shape is a question about a scalar potential.

5.2 The minimal graph

The simplest connected eleven-link graph with one cycle and the observed hierarchies is

$$Y_u = \mathrm{diag}(u_{11}, u_{22}, u_{33}), \qquad m = \begin{pmatrix} m_{11} & m_{12} & 0 \\ 0 & m_{22} & m_{23} \\ 0 & 0 & m_{33} \end{pmatrix}, \qquad B = (0,\, b,\, b\, e^{i\varphi}), \tag{23}$$

plus μ. With Y_u diagonal the whole CKM matrix comes from H_{eff}, and the six parameters $(m_{11}, m_{12}, m_{22}, m_{23}, m_{33}, b/\mu)$ are fixed by the three down-type masses and $|V_{us}|, |V_{cb}|, |V_{ub}|$ (at M_Z). These equations have two exact solutions. One gives

$$(\alpha, \beta, \gamma) = (90.11^\circ,\, 23.89^\circ,\, 65.99^\circ) \qquad \text{at } \varphi = \pi/2, \tag{24}$$

with $|B|/\mu = 2.0$, the order-one ratio that Sec. 4.2 needs and that the fit was not asked for; the other gives $(43.5^\circ, 16.2^\circ, 120.3^\circ)$ and is excluded by the measured α and γ. Following both as φ varies (Fig. 4), the first obeys $\alpha = \varphi$ to within 0.11° over $\varphi \in [\pi/8, 5\pi/8]$, so in this texture the VEV phase *is* the angle α, which is the locking mechanism of Ref. [1] realized by a scalar potential.

Against the data, Eq. (24) pulls by -0.3σ in α, -0.1σ in γ and $+2.6\sigma$ in β. The last number is not the graph's doing. For any triangle the law of sines gives

$$\sin\beta = R_b \sin\alpha, \qquad R_b = \left| \frac{V_{ud} V_{ub}}{V_{cd} V_{cb}} \right|, \tag{25}$$

so once a texture fixes $\alpha = 90^\circ$ and reproduces $|V_{ub}|$ and $|V_{cb}|$, the value $\beta = \arcsin R_b$ is forced, and it is also the largest β available at these magnitudes (Fig. 5). With the average $|V_{ub}| = 3.82 \times 10^{-3}$ this is 23.89°; with the exclusive 3.70×10^{-3} it is 23.10°, a 1.0σ pull; with 3.60×10^{-3} it is 22.44°. The tension is the long-standing $|V_{ub}|$ puzzle [40, 44, 45] seen through the triangle. LHCb is projected to measure β to 0.2° with 50 fb^{-1} and 0.08° with 300 fb^{-1} [1], so together with a settled $|V_{ub}|$ the relation $\alpha = \pi/2$ will be tested sharply.

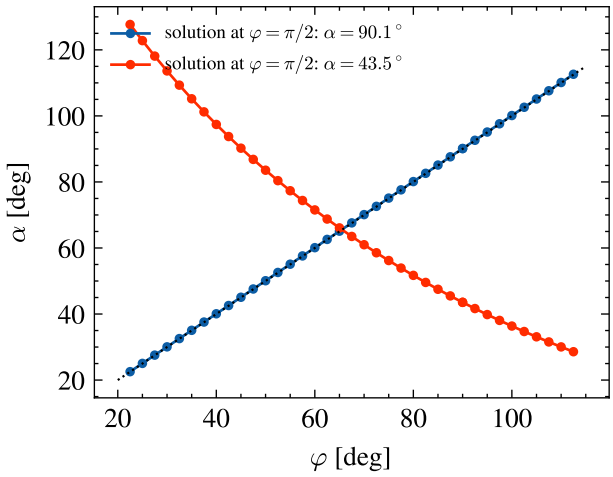


Figure 4: The angle α for the two exact solutions of the minimal graph, Eq. (23), followed as the bridge phase φ varies. One branch tracks $\alpha = \varphi$ (dotted) to within 0.11°.

5.3 All eleven-link graphs

Equation (23) was chosen by hand, so we enumerated the whole family. The admissible links are q_i–u^c_j, q_i–d^c_j, the bridges D_L–d^c_j and the mandatory D_L–D^c, with the q_i–D^c links absent. Requiring that Y_u and m each have a perfect matching (no massless quark), that the graph be connected with eleven links, and that it contain at least two bridges, the graph version of $n_\chi \geq 2$, leaves 6696 graphs (26,352 without the bridge condition), which fall into 34 classes under $S_3^{(q)} \times S_3^{(u^c)} \times S_3^{(d^c)}$: $(n_u, n_m, n_B) = (4,4,2)$ for 16, $(3,5,2)$ and $(5,3,2)$ for 8 each, and $(3,4,3)$, $(4,3,3)$ for one each. A third bridge spends a link that the two matchings and connectivity would otherwise need, which is why so few classes have one.

For each class we solved the nine magnitude equations (six masses and $|V_{us}|, |V_{cb}|, |V_{ub}|$ at M_Z) at $\varphi = \pi/2$ with equal bridge moduli, from several hundred random starting points, and kept every distinct exact solution; since magnitudes cannot tell V from V^*, solutions are folded onto $J > 0$. Twelve classes have no exact solution,[1] five have only CP-conserving ones, and seventeen have CP-violating solutions, usually more than one. Across the family these solutions land on just four points in the (α, β, γ) plane (Fig. 6):

	α	β	γ
A (11 classes)	88.7°–90.1°	23.89°	66.0°–67.4°
B (5 classes)	66.9°–68.0°	21.9°–22.1°	90.0°–91.2°
C (4 classes)	60.9°–62.0°	20.7°–21.0°	97.1°–98.3°
D (4 classes)	43.5°–44.1°	16.2°–16.4°	119.6°–120.3°

On branch A the phase locks α, on branch B it locks γ, and on C and D it locks neither. Only A is compatible with the measured α and γ; B misses α by 8σ, and C and D by more. The minimal graph belongs to A and is its best fit. The robust statement is therefore that every eleven-link texture consistent with the measured angles has $\alpha \simeq \varphi$, with β then given by Eq. (25).

[1] Two of them come within a relative 10^{-4} of one, on branches A and B below; at the precision of the input masses they would count as fits, and they do not change any conclusion.

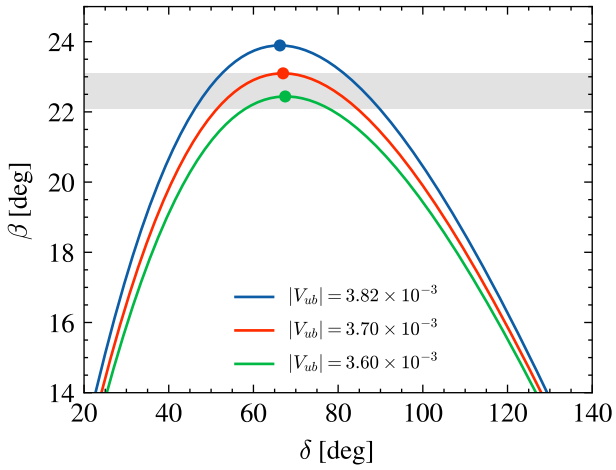


Figure 5: β as a function of the CKM phase δ at fixed $|V_{us}|$, $|V_{cb}|$ and three values of $|V_{ub}|$. Dots mark $\alpha = 90°$, which coincides with the maximum of each curve, Eq. (25). The band is the measured β.

5.4 Without the vector-like quark

Removing D_L and D^c leaves nine nodes and nine links, exactly the setting of Ref. [1]. Handing our fitting code their Example I, with Y_u nonzero at (12), (21), (22), (33), m nonzero at (12), (21), (22), (23), (33) and the phase $\pi/2$ on Y^u_{12}, returns $\alpha = 88.92°$, $\beta = 23.89°$ and $\gamma = 67.19°$, inside their $\varphi = \pi/2$ cluster $[88.74°, 90.03°]$ and between their classes at 88.914° and 88.957°, together with a second exact solution near branch D. Its β is again $\arcsin R_b$: their region I sits at $\beta \simeq 23.65°$–$23.95°$, while their $\varphi = \pi/8$ classes reach $\beta \simeq 22.5°$ at $\delta \simeq 85°$, a different point on the same curve (Fig. 5), possible because those classes do not fit all three magnitudes at once. The two frameworks overlap in α: our branch A spans 88.7°–90.1°, just reaching past the upper edge of their $\varphi = \pi/2$ islands, and the nearest of their classes, with $\alpha = 90.028°$, $\beta = 22.683°$ and $\gamma = 66.253°$, differs from Eq. (24) by 1.2° in β. What separates them is how each treats $|V_{ub}|$, and the comparison should be revisited once that determination settles.

Without D, the parity has nothing to do. Every one of the eighteen SM Yukawa links q–u^c and q–d^c is P_R-even; no node is isolated, the reality of $\det Y$ has to be checked texture by texture, the Nelson–Barr block disappears, and the CP phase must sit in a Yukawa coupling, where ultraviolet CP no longer protects $\bar\theta$. The vector-like quark is what makes the parity bite, what protects $\bar\theta$, and what moves the textures off the nine-link islands. Dense textures, in turn, predict nothing: with the Froggatt–Nielsen matrices of Sec. 4.2 and the same bridge at $\varphi = \pi/2$, the central 68% of α spans 1°–176°, in line with the finding of Ref. [1] that at least four zeros are needed for a robust angle–phase relation. The parity supplies one of them; the rest of Eq. (23) is input.

5.5 Why $\pi/2$, and why not $\pi/8$

The phases of the χ_R VEVs are fixed by the χ sector of the potential, and here the gauge charges force a term that the discussion so far has not used: χ_R carries exactly the quantum numbers of the right-handed lepton doublet, so the scalar analogue of the seesaw coupling,

$$h_{ab}\, \chi_R^{(a)T}\, i\sigma_2\, \Delta_R\, \chi_R^{(b)} + \text{h.c.}, \tag{26}$$

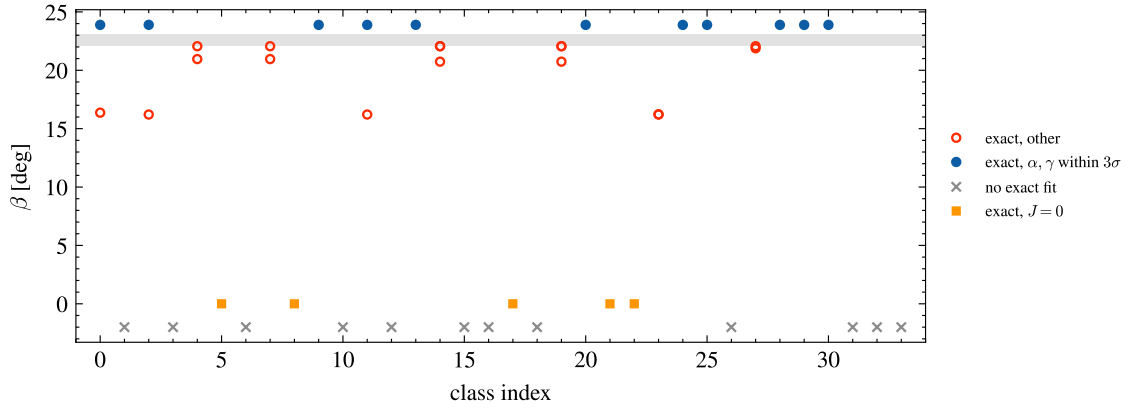


Figure 6: Exact solutions of the nine magnitude equations for each of the 34 eleven-link classes at $\varphi = \pi/2$. A class can have several solutions. Filled circles have α and γ within 3σ of the data; open circles do not; squares are CP-conserving solutions ($\beta = 0$); crosses mark classes with no exact solution. The band is the measured β.

is allowed by every symmetry of the model ($B-L$: $-1-1+2=0$; center: $(-)(-)(+)=+$). It is cubic, and with $\langle \Delta_R \rangle = v_R$ it contributes $v_R w^2 [h_{11}\cos 2\theta_1 + h_{22}\cos 2\theta_2 + 2h_{12}\cos(\theta_1+\theta_2)]$, which dominates the phase-dependent quartics $m_{12}^2 w^2 \cos\varphi$ and $\lambda_5 w^4 \cos 2\varphi$ by $v_R/w \sim 10^6$ unless h_{ab} is absurdly small. The phases are therefore set by Eq. (26), and the outcome depends on the symmetry of the χ sector:

- Under a plain exchange $\chi_1 \leftrightarrow \chi_2$, $h_{11} = h_{22}$, and the minimum has $\varphi \in \{0, \pi\}$: the two VEVs are relatively real and the model has no CP violation at all. The exchange symmetry that makes $|B_2| = |B_3|$ natural is therefore not an option by itself.

- Under the twisted exchange $E: \ \chi_1 \to -i\chi_2, \ \chi_2 \to -i\chi_1$, a $\mathbb{Z}_4$ with $E^2 : \chi_a \to -\chi_a$, one has $h_{22} = -h_{11}$ and $h_{12} = 0$, while $|\chi_1|^2 \leftrightarrow |\chi_2|^2$ still forces $w_1 = w_2$. The cubic term becomes $-2h\, v_R w^2 \sin(\theta_1+\theta_2)\sin(\theta_1-\theta_2)$, whose minimization in $\theta_1-\theta_2$ gives $\cos\varphi = 0$ exactly, for either sign of h and independently of $\theta_1+\theta_2$:
$$\varphi = \pm\frac{\pi}{2}, \tag{27}$$
shifted only by $m_{12}^2/(2hv_R) \lesssim 10^{-6}$ from the subleading quartics. The two signs are the CP-conjugate vacua of Sec. 5.3.

- With no symmetry, $h_{11}h_{22} < 0$ gives $|\varphi|$ within a few degrees of $\pi/2$ and $h_{11}h_{22} > 0$ gives $\varphi \in \{0, \pi\}$.

A $\mathbb{Z}_4$ acting on the two phase words thus yields $\varphi = \pi/2$ from the potential itself, with the $\mathbb{Z}_4$ root of unity entering through $E^2 = -1$ rather than through a larger gauge group; the Pati–Salam $\mathrm{SU}(4)_C$ center remains a possible gauge origin for E but is not needed for the result. E is a symmetry of the scalar potential; the bridge couplings break it, as they must, since an E-invariant bridge would give every B_j the same phase. The bridges couple χ_1 and χ_2 to orthogonal flavor directions, so they do not regenerate h_{12} or the $\chi_1^\dagger\chi_2$ mass mixing at one loop, and a mild violation of E shifts φ by a small amount that does not touch $\bar\theta$, which is protected by the gauge parity alone.

The same logic stops short of $\beta \simeq \pi/8$, which Ref. [1] obtains from a $\mathbb{Z}_8$ acting on flavons through

$$\frac{z_1 - z_2}{z_1 + z_2} = -i\tan\frac{\theta}{2}, \qquad |z_1| = |z_2|, \quad \theta = \arg(z_2/z_1), \tag{28}$$

since no cubic or quartic invariant of two doublets quantizes a relative phase to $\pi/8$. The potential accounts for $\alpha \simeq \pi/2$; β is left to $|V_{ub}/V_{cb}|$.

5.6 An axion instead?

The mechanism needs no axion, but the structure tolerates one. A Peccei–Quinn axion at $f_a \sim 10^{15}$ GeV would need PQ-violating operators suppressed up to dimension $d \gtrsim 23$ for order-one Planck-normalized coefficients, and the center parity removes only half of them. Its mass would be

$$m_a \simeq \frac{f_\pi m_\pi}{f_a} \frac{\sqrt{m_u m_d}}{m_u + m_d} \approx 5.7 \text{ neV} \, \frac{10^{15} \text{ GeV}}{f_a}, \tag{29}$$

in the range of DMRadio and CASPEr [46,47]. The two options are cleanly distinguishable: the Nelson–Barr route predicts no axion and a CP-breaking sector at or below 10^9 GeV.

6 Conclusions

The discrete symmetry that Nelson–Barr models introduce by hand can be part of the gauge group, and the SM hypercharges point to which one. In the left-right completion, an $\mathrm{SU}(2)_R$-singlet vector-like quark cannot couple to the quark doublets through the bidoublet, the Bento–Branco–Parada texture follows, and $\arg\det\mathcal{M} = 0$ holds for every texture while the CKM phase stays generic: over 400 random textures the measured Jarlskog invariant lands at the 30th percentile. The protecting symmetry is the center of $\mathrm{SU}(2)_R$, which survives the triplet that breaks $\mathrm{SU}(2)_R$ and sets the seesaw scale, and which below that scale is matter parity. Its quality is fixed by a dimension-five operator and puts the CP-breaking sector below 10^9 GeV, lower if the ultraviolet does not respect flavor. The bidoublet vacuum is relatively real to the required accuracy as a consequence of the potential: its only CP-odd tadpole is a portal to the two CP-breaking doublets, suppressed by $(\langle\chi_R\rangle/v_R)^2 \sim 10^{-12}$–$10^{-14}$, which bounds the CP-breaking scale by the same 10^9 GeV as the quality argument. The cubic coupling of the doublets to the triplet fixes their relative phase, and a $\mathbb{Z}_4$ exchange symmetry of the potential sets it to $\pi/2$ exactly.

In the language of link diagrams the gauge zero removes every phase-carrying perfect matching at once, and the one physical phase becomes the relative phase of two VEVs. Among all 6696 eleven-link graphs, the solutions compatible with the measured angles lie on a single branch on which that phase is the angle α. The value $\varphi = \pi/2$ selected by the $\mathbb{Z}_4$ then gives $(90°, 23.9°, 66°)$, with β determined by R_b, so the construction will be tested by the LHCb measurement of β together with the resolution of $|V_{ub}|$, and, independently of flavor, by the absence of an axion.

Acknowledgements

D.Š. and D.S. are partially supported by the US National Science Foundation under Grant No. PHY-2310363.

Disclosure of generative AI use Claude (Anthropic) was used to improve the text and to assist in writing the numerical code. The authors have verified all derivations, numerical results and references, and take full responsibility for the content.

Author contributions Both authors contributed equally to all aspects of this work.

Funding information US National Science Foundation, Grant No. PHY-2310363.

Code availability The Python code that produces every number and figure in this paper is provided with the submission.

A Registers in components

With $|1\rangle, |2\rangle, |3\rangle = |r\rangle, |g\rangle, |b\rangle$, the color generators in matrix units are

$$\begin{aligned} &\lambda_1 = e_{12} + e_{21}, \quad &&\lambda_2 = i(e_{21} - e_{12}), \quad &&\lambda_3 = e_{11} - e_{22}, \quad &&\lambda_4 = e_{13} + e_{31}, \\ &\lambda_5 = i(e_{31} - e_{13}), \quad &&\lambda_6 = e_{23} + e_{32}, \quad &&\lambda_7 = i(e_{32} - e_{23}), \quad &&\lambda_8 = \tfrac{1}{\sqrt{3}}(e_{11} + e_{22} - 2e_{33}). \end{aligned} \tag{A.1}$$

The raising operators e_{12}, e_{13}, e_{23} and their conjugates are the six charged gluon vertices; e_{12}, for instance, turns a green quark red. λ_3 weighs r against g and λ_8 weighs r, g against b, the two Cartan generators of a rank-two algebra.

In the same language u_R carries $n_1 = +4$ unary indices, e_R carries six anti-indices and ν_R none. On the binary state $|0110\rangle = |d_L u_L u_L d_L\rangle$, σ^+ acting on the first slot gives $|1110\rangle$ and on the second gives zero, the nilpotency of the raising operator in the fundamental representation; a W^+ vertex is such a single-slot substitution. On $|132\rangle = |rbg\rangle$, e_{23} acting on the second slot gives $|rgg\rangle$. Of the six orderings of rgb, only $\frac{1}{\sqrt{6}}\varepsilon_{ijk}|ijk\rangle$ is invariant, picking up $\det U = 1$: color neutrality belongs to the combination, not to any single ordering.

B Numerical procedure

All numbers and figures are produced by the short Python scripts provided with the submission. The inputs are $\overline{\text{MS}}$ masses at M_Z and CKM magnitudes [40],

$$(m_u, m_c, m_t) = (0.0013,\, 0.62,\, 172)\ \text{GeV}, \quad (m_d, m_s, m_b) = (0.0028,\, 0.055,\, 2.89)\ \text{GeV}, \tag{B.1}$$

with $|V_{us}| = 0.2243$, $|V_{cb}| = 0.0410$ and $|V_{ub}| = 0.00382$.

Random textures. Real matrices $m_{ij} = c_{ij}\lambda^{a_i + a_j}$ with $a = (2, 1, 0)$ (down) and $(4, 2, 0)$ (up), c_{ij} of random sign with $|c_{ij}|$ log-uniform in $[\frac{1}{2}, 2]$ (CP requires real, not positive, couplings), are scaled so that their largest singular values are 2.9 and 173 GeV. The bridge is $B_i \propto r_i + ir'_i$ with standard normal r_i, r'_i and $|B| = \mu$; for the one-doublet control all B_i share a phase. The CKM matrix follows from diagonalizing H_{eff} and $m_u m_u^\dagger$. Median singular values are $(0.011, 0.18, 2.9)$ GeV for m and $(0.0019, 0.47, 173)$ GeV for m_u, within a factor of four of the measured masses. The coefficient c of Eq. (11) is evaluated on the same textures as $\text{Tr}(M_d^{-1} M_u) - \text{Tr}(M_u^{-1} M_d)$; it agrees with the finite-difference derivative of $\arg\det(M_u M_d)$ with respect to $\text{Im}\, z$ once the $\bar{\theta} = \pi$ branch of sign-flipped determinants is removed, and has median 45 and 10–90% range 9 to 300.

Leakage. For Fig. 2, μ is log-uniform in $[10^4, 10^{13}]$ GeV, F log-uniform in $[0.1, 1]$ and $|\langle \chi_R \rangle| = \mu/F$, with two doublets of random relative phase. The dimension-five entry is $\Delta = m\, k\, \langle\chi\rangle^* / M_{\text{Pl}}$ in the aligned case and $\Delta = m_b\, k\, \langle\chi\rangle^* / M_{\text{Pl}}$ in the anarchic one, with k standard normal. The phase is computed from the Schur complement $\mu - B m^{-1} \Delta$, since a

direct 4×4 determinant loses all precision at $\mu/m_d \sim 10^{15}$. The median of $|\delta\bar{\theta}|\, M_{\rm Pl}\mu/\langle\chi_R\rangle^2$ is 0.18 (aligned) and 22 (anarchic).

Potential minimization. For Eq. (13) the full bidoublet potential, including all quartics and the portals (12), is minimized in (κ_1, X, Y) with 60-digit arithmetic, the light eigenvalue of the (κ_1, κ_2) mass matrix tuned to $-(100\ \text{GeV})^2$, for random couplings with μ_2^2 of order v_R^2; the Hessian is checked to be positive. For Eq. (27) the χ-sector phase potential, including the cubic term (26), m_{12}^2, λ_5 and λ_6, is minimized from random starts under each symmetry choice.

Graph scan. `scan.py` enumerates the graphs, reduces them to classes by brute-force canonical forms under S_3^3, and for each class runs bounded least squares on the logarithms of the nine observables from 500 random starts. A solution is kept if the sum of squared log-residuals is below 10^{-16}, and solutions are grouped by their angles to 0.1°. The phase sits on a bridge on the graph's cycle; if no bridge lies on the cycle, the phase is unphysical and the solutions are CP-conserving.

References

[1] N. Arkani-Hamed, C. Figueiredo, L. J. Hall and C. A. Manzari, *The Very Nearly Right Theory of Flavor* (2026), 2607.27315.

[2] C. Abel, S. Afach, N. J. Ayres, C. A. Baker, G. Ban, G. Bison, K. Bodek, V. Bondar, M. Burghoff, E. Chanel, Z. Chowdhuri, P.-J. Chiu *et al.*, *Measurement of the permanent electric dipole moment of the neutron*, Phys. Rev. Lett. **124**, 081803 (2020), 10.1103/PhysRevLett.124.081803.

[3] R. D. Peccei and H. R. Quinn, CP *conservation in the presence of pseudoparticles*, Phys. Rev. Lett. **38**, 1440 (1977), 10.1103/PhysRevLett.38.1440.

[4] R. D. Peccei and H. R. Quinn, *Constraints imposed by* CP *conservation in the presence of pseudoparticles*, Phys. Rev. D **16**, 1791 (1977), 10.1103/PhysRevD.16.1791.

[5] M. Kamionkowski and J. March-Russell, *Planck scale physics and the Peccei-Quinn mechanism*, Phys. Lett. B **282**, 137 (1992), 10.1016/0370-2693(92)90492-M, hep-th/9202003.

[6] R. Holman, S. D. H. Hsu, T. W. Kephart, E. W. Kolb, R. Watkins and L. M. Widrow, *Solutions to the strong CP problem in a world with gravity*, Phys. Lett. B **282**, 132 (1992), 10.1016/0370-2693(92)90491-L, hep-ph/9203206.

[7] S. M. Barr and D. Seckel, *Planck scale corrections to axion models*, Phys. Rev. D **46**, 539 (1992), 10.1103/PhysRevD.46.539.

[8] A. E. Nelson, *Naturally weak cp violation*, Physics Letters B **136**, 387 (1984).

[9] S. M. Barr, *Solving the Strong CP Problem Without the Peccei-Quinn Symmetry*, Phys. Rev. Lett. **53**, 329 (1984), 10.1103/PhysRevLett.53.329.

[10] L. Bento, G. C. Branco and P. A. Parada, *A minimal model with natural suppression of strong cp violation*, Physics Letters B **267**(1), 95 (1991), 10.1016/0370-2693(91)90530-4.

[11] M. Dine and P. Draper, *Challenges for the Nelson-Barr Mechanism*, JHEP **08**, 132 (2015), 10.1007/JHEP08(2015)132, 1506.05433.

[12] P. Asadi, S. Homiller, Q. Lu and M. Reece, *Chiral nelson-barr models: Quality and cosmology*, Phys. Rev. D **107**, 115012 (2023), 10.1103/PhysRevD.107.115012.

[13] P. F. Perez, C. Murgui and M. B. Wise, *Automatic Nelson-Barr solutions to the strong CP puzzle*, Phys. Rev. D **108**(1), 015010 (2023), 10.1103/PhysRevD.108.015010, 2302.06620.

[14] C. Murgui and S. Patrone, *Leptogenesis in automatic Nelson-Barr models*, JHEP **09**, 113 (2025), 10.1007/JHEP09(2025)113, 2506.18963.

[15] C. Murgui, *A Pati-Salam realization of the Nelson-Barr mechanism*, 10.1103/9xq1-vym3 (2026), 2604.00166.

[16] K. S. Babu and R. N. Mohapatra, *A Solution to the Strong CP Problem Without an Axion*, Phys. Rev. D **41**, 1286 (1990), 10.1103/PhysRevD.41.1286.

[17] L. J. Hall and K. Harigaya, *Implications of Higgs Discovery for the Strong CP Problem and Unification*, JHEP **10**, 130 (2018), 10.1007/JHEP10(2018)130, 1803.08119.

[18] J. N. Benabou, A. Hook, C. A. Manzari, H. Murayama and B. R. Safdi, *Clearing up the Strong CP problem* (2025), 2510.18951.

[19] J. C. Pati and A. Salam, *Lepton number as the fourth "color"*, Phys. Rev. D **10**, 275 (1974), 10.1103/PhysRevD.10.275.

[20] R. N. Mohapatra and J. C. Pati, *A Natural Left-Right Symmetry*, Phys. Rev. D **11**, 2558 (1975), 10.1103/PhysRevD.11.2558.

[21] G. Senjanovic and R. N. Mohapatra, *Exact left-right symmetry and spontaneous violation of parity*, Phys. Rev. D **12**, 1502 (1975), 10.1103/PhysRevD.12.1502.

[22] L. M. Krauss and F. Wilczek, *Discrete gauge symmetry in continuum theories*, Phys. Rev. Lett. **62**, 1221 (1989), 10.1103/PhysRevLett.62.1221.

[23] T. Banks and N. Seiberg, *Symmetries and Strings in Field Theory and Gravity*, Phys. Rev. D **83**, 084019 (2011), 10.1103/PhysRevD.83.084019, 1011.5120.

[24] L. E. Ibanez and G. G. Ross, *Discrete gauge symmetry anomalies*, Phys. Lett. B **260**, 291 (1991), 10.1016/0370-2693(91)91614-2.

[25] S. P. Martin, *Some simple criteria for gauged R-parity*, Phys. Rev. D **46**, R2769 (1992), 10.1103/PhysRevD.46.R2769, hep-ph/9207218.

[26] S. Powers and D. Stojkovic, *An alternative formalism for modeling spin*, Eur. Phys. J. C **82**, 690 (2022), 10.1140/epjc/s10052-022-10652-y, 2110.13617.

[27] S. Powers, G. Xu, H. Fotso, T. Thomay and D. Stojkovic, *Statistical model for quantum spin and photon number states*, Phys. Rev. A **111**(1), 012217 (2025), 10.1103/PhysRevA.111.012217, 2304.13535.

[28] S. Powers, *A New Quantum Ontology: A Statistical Theory of Event Networks Built from Classical Bits*, Ph.D. thesis, University at Buffalo (2023).

[29] S. Powers and D. Stojkovic, *An event centric approach to modeling quantum systems* (2023), 2306.14922.

[30] P. Berglund *et al.*, *Quantum Gravity and Phenomenology: Dark Matter, Dark Energy, Vacuum Selection, Emergent Spacetime, and Wormholes*, In *Snowmass 2021* (2022), 2202.05104.

[31] S. Powers, D. Škrgić and D. Stojkovic, *The quantum harmonic oscillator from binary sequences* (2026), 2609.10568.

[32] S. Powers and D. Stojkovic, *Emergent Space, Time, and Lorentz Symmetry from Binary Sequences* (2026), 2608.23586.

[33] H. Georgi, *Lie Algebras In Particle Physics: from Isospin To Unified Theories*, Frontiers in Physics. Avalon Publishing, ISBN 9780813346113 (1999).

[34] G. 't Hooft, *On the phase transition towards permanent quark confinement*, Nuclear Physics B **138**(1), 1 (1978), 10.1016/0550-3213(78)90153-0.

[35] A. M. Polyakov, *Thermal Properties of Gauge Fields and Quark Liberation*, Phys. Lett. B **72**, 477 (1978), 10.1016/0370-2693(78)90737-2.

[36] J. Greensite, *The Confinement problem in lattice gauge theory*, Prog. Part. Nucl. Phys. **51**, 1 (2003), 10.1016/S0146-6410(03)90012-3, hep-lat/0301023.

[37] D. Tong, *Line Operators in the Standard Model*, JHEP **07**, 104 (2017), 10.1007/JHEP07(2017)104, 1705.01853.

[38] P. Pfeuty, *The one-dimensional ising model with a transverse field*, Annals of Physics **57**, 79 (1970).

[39] A. Valenti and L. Vecchi, *The CKM phase and $\bar{\theta}$ in Nelson-Barr models*, Journal of High Energy Physics **2021**(7), 203 (2021), 10.1007/JHEP07(2021)203, 2105.09122.

[40] S. Navas *et al.*, *Review of particle physics*, Phys. Rev. D **110**(3), 030001 (2024), 10.1103/PhysRevD.110.030001.

[41] T. W. B. Kibble, G. Lazarides and Q. Shafi, *Walls bounded by strings*, Phys. Rev. D **26**, 435 (1982), 10.1103/PhysRevD.26.435.

[42] D. Stojkovic, K. Freese and G. D. Starkman, *Holes in the walls: Primordial black holes as a solution to the cosmological domain wall problem*, Phys. Rev. D **72**, 045012 (2005), 10.1103/PhysRevD.72.045012, hep-ph/0505026.

[43] M. Bona *et al.*, *New UTfit Analysis of the Unitarity Triangle in the Cabibbo-Kobayashi-Maskawa scheme*, Rend. Lincei Sci. Fis. Nat. **34**, 37 (2023), 10.1007/s12210-023-01137-5, 2212.03894.

[44] F. U. Bernlochner, *Semileptonic decays and the determination of $|V_{ub}|$*, In *9th Conference on Flavor Physics and CP Violation* (2011), 1110.1325.

[45] L. Cao *et al.*, *First Simultaneous Determination of Inclusive and Exclusive —Vub—*, Phys. Rev. Lett. **131**(21), 211801 (2023), 10.1103/PhysRevLett.131.211801, 2303.17309.

[46] L. Brouwer, S. Chaudhuri, H.-M. Cho, J. Corbin, W. Craddock, C. S. Dawson, A. Droster, J. W. Foster, J. T. Fry, P. W. Graham, R. Henning, K. D. Irwin *et al.*, *Projected sensitivity of dmradio-m^3: A search for the qcd axion below* 1 μeV, Phys. Rev. D **106**, 103008 (2022), 10.1103/PhysRevD.106.103008.

[47] D. Budker, P. W. Graham, M. Ledbetter, S. Rajendran and A. O. Sushkov, *Proposal for a cosmic axion spin precession experiment (casper)*, Phys. Rev. X **4**, 021030 (2014), 10.1103/PhysRevX.4.021030.